

Tianwei Ye

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Education

- National University of Singapore**, School of Computing 2026.8 – Present
Ph.D. Student, supervised by Prof. Gim Hee Lee
- **Research direction:** 3D vision and embodied AI
- Wuhan University**, Multi-Spectral Vision Processing Lab 2024.9 – 2026.6
M.Sc in Electronic Information, supervised by Prof. Jiayi Ma and Prof. Yong Ma
- **GPA:** 3.62/4.00
 - **Courses:** Advanced Engineering Mathematics (A+), Modern Circuit Design (A), Stochastic Processes (A-), Optimization Theory and Methods (A+)
- Central South University**, School of Electronic Information 2020.9 – 2024.6
B.Sc in Electronic Information Science and Technology
- **GPA:** 91.25/100, ranking top 7% out of 150
 - **Courses:** Digital Electronic Technology (94), Signals and Systems (90), Microcontroller Systems and Interface Technology (96), Algorithms and Data Structures (93), Digital Signal Processing (99)

Publications

- SGMatch: Semantic-Guided Non-Rigid Shape Matching with Flow Regularization** [PDF] [Code]
Tianwei Ye, Xiaoguang Mei, Yifan Xia, Fan Fan, Jun Huang, Jiayi Ma 2026
- Proc. of the European Conference on Computer Vision (ECCV), 2026
 - Architected a semantic-geometric fusion pipeline featuring a Semantic-Guided Local Cross-Attention module for symmetry disambiguation and a Conditional Flow Matching objective to regularize spatially smooth feature transport.
 - Dominated non-isometric and topologically noisy benchmarks, surpassing state-of-the-art results.
- DcMatch: Unsupervised Multi-Shape Matching with Dual-Level Consistency** [PDF] [Code]
Tianwei Ye, Yong Ma, Xiaoguang Mei 2026
- Proc. of the AAAI Conference on Artificial Intelligence (AAAI), 2026
 - Engineered a novel unsupervised framework utilizing a shape graph attention module to extract manifold-aware features from entire collections, establishing a robust shared universe space with dual-level (spectral and spatial) cycle consistency.
- Probabilistic Deformation Consistency for Unsupervised Shape Matching** [PDF] [Code]
Yifan Xia, **Tianwei Ye**, Jun Huang, Xiaoguang Mei, Jiayi Ma 2026
- Proc. of the AAAI Conference on Artificial Intelligence (AAAI), 2026
 - Developed a non-Euclidean probabilistic deformation framework that leverages a linear Expectation-Maximization procedure and spectral kernel techniques to jointly estimate smooth deformation fields and correspondence reliability.
- Multi-Shape Matching with Cycle Consistency Basis via Functional Maps** [PDF] [Code]
Yifan Xia*, **Tianwei Ye***, Huabing Zhou, Zhongyuan Wang, Jiayi Ma 2025
- Proc. of the AAAI Conference on Artificial Intelligence (AAAI), 2025
 - Developed a two-stage optimization framework modeling shape collections as directed graphs, utilizing Local Mapping Distortion for landmark-driven accuracy and a novel cycle consistency basis to succinctly enforce global constraints via ADMM.

Preprints

3D Generation for Embodied AI and Robotic Simulation: A Survey [PDF] [Website]

Tianwei Ye, Yifan Mao, Minwen Liao, Jian Liu, Chunchao Guo, Dazhao Du, 2026
Quanxin Shou, Fangqi Zhu, Song Guo

- Systematized 3D generation for embodied AI around three functional pillars: simulation-ready asset generation (articulated, physical, and deformable objects), interactive scene generation (controllable and agentic environments), and Sim2Real transfer (digital twins, data augmentation, and synthetic demonstrations for robot learning).
- Identified the field's shift from visual realism to interaction readiness, highlighting critical bottlenecks including limited physical annotations, the geometric-physical validity gap, fragmented evaluation, and the persistent sim-to-real divide.

Research Experience

3D World Model for Embodied AI

Remote Research Intern, PEI-lab HKUST, supervised by Prof. Song Guo 2025.11 – 2026.05

- Conducted a comprehensive survey on 3D generative models for Embodied AI, covering object-level synthesis, scene-level simulation, and Sim-to-Real transfer.
- Investigated block-diffusion-based Vision-Language-Action models, focusing on cross-modal dependency modeling and few-step generation for real-time inference.

Probabilistic Joint Learning for Multimodal Image Matching

NSFC Basic Research Project, hosted by Yifan Xia 2025.1 – 2025.10

- Developed a probabilistic framework for multimodal image registration (Visible/IR) under non-linear appearance variations and data scarcity, using conditional diffusion models.
- Designed a geometric constraint module to recover transformation fields in large-rotation scenes, improving robustness over standard baselines.

UAV-Based Hyperspectral Imaging Technology for Precise Detection of Marine Pollution

Supervised by Prof. Fan Fan and Prof. Jun Huang 2024.9 – 2026.2

- Built a UAV-based hyperspectral remote sensing system for real-time detection of marine red tides and oil spills.
- Implemented anomaly detection and source localization modules, outperforming traditional baselines on maritime datasets.

Honors & Awards

Second Prize Scholarship, Wuhan University	2025
Second Prize, China Postgraduate Mathematical Contest in Modeling	2025
Outstanding Graduate, Central South University	2024
First Prize Scholarship, Central South University	2023
First Prize, Hunan Division, China Undergraduate Mathematical Contest in Modeling	2022
Merit Student, Central South University	2022
Second Prize Scholarship, Central South University	2021

Skills

Mathematics: 3D Geometry, Linear Algebra, Numerical Optimization

Programming: Python, C/C++, Matlab, $\text{\LaTeX}$

Development Tools: Git, Docker, Linux, Claude Code, Codex

Language: English (*Proficient*, IELTS 7.0), Chinese (*Native*)